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| Course Title                                                                                                                                                                                  | Operating Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |     |
| Code                                                                                                                                                                                          | CS-241                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |
| Credit hours                                                                                                                                                                                  | 2+1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |     |
| Prerequisite                                                                                                                                                                                  | Data Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |     |
| Category                                                                                                                                                                                      | Core                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |     |
| Course Description                                                                                                                                                                            | To help students gain a general understanding of the principles and concepts governing the functions of operating systems and acquaint students with the layered approach that makes design, implementation and operation of the complex OS possible.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |     |
| Course Outline                                                                                                                                                                                | Operating systems basics, system calls, process concept and scheduling, inter-process communication, multithreaded programming, multithreading models, threading issues, process scheduling algorithms, thread scheduling, multiple-processor scheduling, synchronization, critical section, synchronization hardware, synchronization problems, deadlocks, detecting and recovering from deadlocks, memory management, swapping, contiguous memory allocation, segmentation & paging, virtual memory management, demand paging, thrashing, memory-mapped files, file systems, file concept, directory and disk structure, directory implementation, free space management, disk structure and scheduling, swap space management, system protection, virtual machines, operating system security |          |     |
| Aim & Objectives                                                                                                                                                                              | The aim of this course is to provide a comprehensive understanding of Operating Systems, including their design, implementation, and management. The objective is to equip students with the knowledge and skills to design, implement, and manage modern operating systems efficiently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |     |
| Mapping of CLOs with PLOs                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |     |
| Course Learning Outcomes (CLOs)                                                                                                                                                               | Domain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | BT Level | PLO |
| Explain the characteristics of different structures of the Operating Systems and identify the core functions of the Operating Systems                                                         | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2        | 2   |
| Analyze the algorithms of the core functions of the Operating Systems and explain the major performance issues with regard to the core functions                                              | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4        | 3   |
| Demonstrate the knowledge in applying system software and tools available in modern operating systems.                                                                                        | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3        | 5   |
| Mapping of LLOs with PLOs                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |     |
| Lab Learning Outcomes (LLOs)                                                                                                                                                                  | Domain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | BT Level | PLO |
| Explain the instructions to implement database system queries and operations while considering operating systems concepts, including process management, file systems, and memory management. | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2        | 2   |
| Practice the Operating Systems lab task as per requirement using modern tools.                                                                                                                | P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4        | 5   |
| Ability to communicate effectively; orally as well as in writing.                                                                                                                             | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2        | 7   |
| Reference Materials                                                                                                                                                                           | 1. Operating Systems Concepts, 9th edition by Abraham Silberschatz<br>2. Modern Operating Systems, 4th edition by Andrew S. Tanenbaum.<br>3. Operating Systems, Internals and Design Principles, 9th edition by William Stallings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |     |
| Grading Breakup and Policy                                                                                                                                                                    | Quiz 1: 10%<br>Quiz 2: 10%<br>Sessional 10%<br>Midterm Examination: 30%<br>Final Examination: 40%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |     |
| Week#                                                                                                                                                                                         | Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |     |
| 1                                                                                                                                                                                             | Operating Systems Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |     |
| 2                                                                                                                                                                                             | System Calls                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |     |
| 3                                                                                                                                                                                             | Process Concept                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |     |
| 4                                                                                                                                                                                             | Process Scheduling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |     |

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|------------------------|----------------------------------------------------------------------------------------------|
| 5                      | Inter-Process Communication                                                                  |
| 6                      | Multithreaded Programming                                                                    |
| 7                      | Multithreading Models                                                                        |
| 8                      | Threading Issues                                                                             |
| <b>Mid Term Exam</b>   |                                                                                              |
| 9                      | Process Scheduling Algorithms                                                                |
| 10                     | Thread Scheduling                                                                            |
| 11                     | Multiple-Processor Scheduling, Synchronization                                               |
| 12                     | Critical Section and Synchronization Hardware                                                |
| 13                     | Synchronization Problems and Deadlocks                                                       |
| 14                     | Memory Management, Swapping, Contiguous Memory Allocation                                    |
| 15                     | Segmentation, Paging, Virtual Memory Management, File Systems                                |
| 16                     | Directory and Disk Structure, System Protection, Virtual Machines, Operating System Security |
| <b>Final Term Exam</b> |                                                                                              |